Changing sun direction / time of day in architectural images: test task summary

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What I did

I tested 4 tools on 3 photos (old barn with low sun, timber building at sunset, interior with window blinds), 18 runs, free options only: **Sun Direction LoRA** (FLUX-based) and **IC-Light** on Hugging Face Spaces, **Qwen-Image-Edit-2509 Relight LoRA** on ComfyUI Cloud, and the commercial tool **Relight Spaces**. I also planned Relighting Kontext LoRA v3, but its Space was unavailable and returned errors, so it was not tested.

Main finding: no tool both moved the sun direction and kept the scene intact. The only clear move of the light was Sun Direction LoRA on the interior; every other result changed color, sky or brightness while shadows stayed in place.

What I found and what went wrong

Tool	What worked	What went wrong	Time	Cost
Sun Direction LoRA	Only tool that moved the light: interior at 6° elevation, blind stripes kept. Sunset building turned into a midday-like day.	Added a second chimney on the barn; lamps switched on; colors grey; white instead of blue sky; output 57–71% of input width; 36° elevation smeared the blind stripes. Base model non-commercial.	13–17 s	Free Space
Qwen Relight LoRA	Kept geometry, objects and lamps on barn and interior; Apache 2.0.	Did not move the sun (more saturation/contrast only). On the building: mild prompt kept the golden hour and added a chimney, strong prompt invented a second house, strict prompt changed nothing.	5–44 s	Free tier
IC-Light	Fast; building left untouched except the sky.	Did not change the sun; barn wall greyed and darker; Space crops to 4:5 and returns 512×640.	5–7 s	Free Space
Relight Spaces	Best time-of-day change and preservation: sunset became a clear blue-sky day, every object kept, same or larger size.	Fixed presets, no direction control: shadows and light patches stayed in place on all three photos; some detail invented in dark areas.	30–39 s	$0.40–0.75 per photo

How I decided which result is better

Each result is scored 1–5 by eye on four criteria: **Geometry** (shape kept), **Material** (textures/colors), **Light change** (time of day/sun move), and **Shadows**. I back this with numbers against the original: edge/shape matching (88–97%), mean brightness and color shift, and output resolution. Edge metrics alone missed a regenerated scene, so I also visually verified every output. Scores reflect practical judgment on 3 key test cases.

Time and cost per image

Time is what the tool showed or my stopwatch (queue time excluded). Free tools cost $0. Hosted prices come from the providers' sites: Relight Spaces $0.40–0.75 per photo (3 free), Magnific €0.10 per operation and fal IC-Light V2 $0.10 per megapixel (both not tested).

What I would put in production, and how

Time-of-day presets: Run Qwen-Image-Edit-2509 + Relight LoRA on our own GPU (Apache 2.0 open license; FLUX models are non-commercial). Use Relight Spaces' result as the quality benchmark and hosted fallback. Filter bad outputs automatically via size matching, edge overlap, color consistency, and object integrity.

Cost and speed: about 15 s per image typical (up to 44 s). On an assumed $2/hour GPU (illustration, not a quote) that is about $0.008 per image at full utilization and $0.08 at 10%, against $0.10–0.75 hosted. Own GPU wins above roughly 8% utilization (about 20 images per hour); below that a hosted API is cheaper and simpler.

Sun direction itself is R&D: train a LoRA on an Apache-licensed base (FLUX.2 klein 4B or Qwen) with before/after pairs from renders with known sun positions. **Next step:** test Qwen with the trigger word from its model card and a fixed output size.

Limits: 3 photos, 1–3 runs each; free demo GPU limits occasionally interrupted full parameter sweeps.